Juwan Ha

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Research Interest

- Energy Efficiency and Demand Flexibility across Buildings, District Energy Systems, and AI Data Centers
- Physics-Informed Machine Learning and Differentiable Predictive Control
- Scalable and Transferable Coordinated Control of Heterogeneous Buildings and Distributed Energy Resources under Distribution-Network Constraints

Education

08/2022–Present PhD in Design (Design for Energy and Sustainability), North Carolina State University
Supervisor: Soolyeon Cho
Committee: Ning Lu, Jeremiah Johnson, Newton Dsouza
Graduate Certificate in Geographic Information Systems (GIS)

09/2017–08/2019 M.E., Architectural Engineering, Gyeongsang National University
Supervisor: Younghak Song

Thesis: A Study on Circulation Pump Configuration and Cooling Tower Control for Improving the Efficiency of Cooling Source System in Buildings
🏆 ***Outstanding Master's Thesis Award***

03/2011-08/2017 B.S., Architectural Engineering, Gyeongsang National University

Professional Experience

08/2022–Present **Research assistant**, North Carolina State University, Raleigh, NC
Supervisor: Soolyeon Cho

06/2025–08/2025; 06/2026–08/2026 **EEIP PhD Intern**, Pacific Northwest National Laboratory (PNNL), Portland, OR
Supervisor: Michael Poplawski

06/2024–08/2024 **Research Affiliate**, Lawrence Berkeley National Laboratory (LBNL), Berkeley, CA
Supervisor: Donghun Kim

06/2017–08/2022 **Research assistant**, Gyeongsang National University, Jinju, South Korea
Supervisor: Young-hak Song

01/2017–02/2017 **Engineer Intern**, Hyundai Engineering & Construction (Hyundai E&C), Singapore

Honors & Awards

2026 Best Paper in Controls, IBPSA-USA SimBuild

2026 ASHRAE JR Anderson Travel Award

2025, 2026	PhD in Design Travel Award, College of Design, North Carolina State University
2025	Student of the Year Award, The Korean American Society of Civil, Environmental, and Architectural Engineers (KSCEA)
2025	ASHRAE Graduate Student Grant-in-Aid
2025	Sustainable Futures Fellows, North Carolina State University
2025	ASHRAE Student Activities Travel Grant
2025	Linda Latham Scholarship, American Council for an Energy-Efficient Economy (ACEEE) — Awarded, Declined
2024, 2025	Next-Generation Science & Technology Leaders NET, Korean Federation of Science & Technology Societies (KOFST)
2024	The Kenan Institute for Engineering, Technology & Science (KIETS) Climate Leaders Program Fellowship, North Carolina State University
2023, 2024	International Institute of Building Enclosure Consultants (IIBEC) Travel Fellowship
2023	The Association of Energy Engineers (AEE) Foundation Scholarship
2022	University Graduate Fellowships, North Carolina State University
2022	Annual Best Paper Award, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS)
2021	Annual Best Paper Award, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS)
2020	Best Paper Award of Fall Conference**,** Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS)
2020	Outstanding Master's Thesis Award, Architectural Institute of Korea (AIK)
2019, 2020	Gyeongsang National University (GNU) Young Pioneer Researcher Award
2019	Annual Best Paper Award, Architectural Institute of Korea (AIK)

Publications

Google Scholar: https://scholar.google.com/citations?user=OKbm5IMAAAAJ&hl=en&authuser=1

Journal Articles and Manuscripts

[J20] ***Juwan Ha***, Seungmin Lee, Yujin Kim, Soolyeon Cho. "Bridging Prediction and Optimization: Decision-Focused Learning in Grid interactive Efficient Building". Applied Energy (In-Progress)

[J19] ***Juwan Ha***, Seungmin Lee, Yujin Kim, Soolyeon Cho. "Co-Optimization of Energy and Demand Flexibility via Differentiable Predictive Control in Multi-zone Buildings". Applied Energy (In-Progress)

[J18] ***Juwan Ha***, Seungmin Lee, Yujin Kim, Sunho Kim, Soolyeon Cho. "Differentiable Predictive Control for Demand Flexibility in Multi-zone Commercial Buildings". Energy and Buildings (Under review)

[J17] Seungmin Lee, ***Juwan Ha***, Wenyuan Tang, Soolyeon Cho. "Reframing Building–Grid Recovery: From Outcome-Based Metrics to Trajectory-Based Interpretation", Applied Energy. (Under review)

[J16] Seungmin Lee, ***Juwan Ha***, Wenyuan Tang, Soolyeon Cho. "Forecasting Rare, Large-Scale Weather-Induced Power Outages Using Temporal Fusion Transformers: A Multi-County Study for Building Energy Resilience". Science and Technology for the Built Environment. (Under review)

[J15] KwangHee Kim, Yujin Kim, ***Juwan Ha***, Younghak Song. “AI-based Model Predictive Control of Chiller Plant with Thermal Energy Storage Combined a Water-side Economizer System in Data Center”. Energy and Buildings (I.F. 7.1). Aug. 2026, 117668.

[J14] KwangHee Kim, Yujin Kim, ***Juwan Ha***, Younghak Song. “Development of Integrated Loop Control Algorithm for Improvement of Water-side Economizer Efficiency and Expansion”, International Journal of Refrigeration (I.F. 3.8). Sep. 2025.

[J13] Yujin Kim, KwangHee Kim, ***Juwan Ha***, Younghak Song. “Research on a Plan of Free Cooling Operation Control for the Efficiency Improvement of Water-side Economizer”, Energies (I.F. 3.2), 2024, 17(12), 2804.

[J12] ***Juwan Ha***, Yujin Kim, Younghak Song. “Energy Saving Evaluation with Low Liquid to Gas Ratio Operation in HVAC&R system”, Energies (I.F. 3.252), 2022, 15, 7327.

[J11] Yujin Kim, ***Juwan Ha***, Younghak Song. “Energy Efficiency Analysis on Free Cooling Operation Mode with Water-side Economizer System in Data Center”, Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol. 16, No. 1, pp. 1-13, Feb. 2022.

[J10] ***Juwan Ha*****,** Yujin Kim, Hwanyong Kim, Younghak Song. “Development of Control Algorithm for Cooling Tower Low-Approach Temperature”, Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol. 15, No. 5, pp.431-442, Oct. 2021.

[J9] Seongcheol Park, ***Juwan Ha***, Kyungsoon Park, Younghak Song. “Sensitivity Analysis and Bayesian MCMC based on Artificial Neural Network Surrogate Model for Input Variable Uncertainty Calibration”, Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol. 15, No. 4, pp.326-337, Aug. 2021.
Annual Best Paper

[J8] Yujin Kim, ***Juwan Ha***, Kyungsoon Park, Younghak Song. “A Study on the Energy Reduction Measures of Data Centers through Chilled Water Temperature Control and Water-side Economizer”, Energies (I.F. 3.004), 2021, 14(12), 3575.

[J7] Yujin Kim, ***Juwan Ha***, Kyungsoon Park, Keonho Lee, Younghak Song. “Energy Evaluation of Data Center with Low Cooling Water Temperature and Free Cooling”, Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol. 14, No. 5, pp.427-438, Oct. 2020.
Annual Best Paper

[J6] ***Juwan Ha***, Yujin Kim, Hwanyong Kim, Younghak Song. “Verification of Low-temperature Condenser Water and Operation Time considering Climate Zones”, Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol. 14, No. 3, pp.298-309, Jun. 2020.

[J5] ***Juwan Ha***, Honghee Chang, Kwangjun Cha, Younghak Song. “Verification of Thermal Environment and THI Prediction Equation of Hen House by Field Measurement and Model-based Simulation”, Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol. 14, No. 2, pp.208-219, Apr. 2020.

[J4] ***Juwan Ha***, Soolyeon Cho, Hwanyong Kim, Younghak Song. “Annual Energy Consumption Cut-off with Cooling System Design Parameter Changes in Large Office Buildings”, Energies (I.F. 2.707), 2020, 13(8), 2034.

[J3] ***Juwan Ha***, Kyungsoon Park, Hwanyong Kim, Younghak Song. "A Study on Annual Cooling Energy Consumption Cut-off with Cooling Tower Control Change", Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol.13, No.6, pp.503-514, Dec. 2019

[J2] Seongcheol Park, ***Juwan Ha***, Hwanyong Kim, Younghak Song. "An Operation Status of Library Building using BEMS Data -Energy Performance Evaluation on Initial Stage of Completion-", Journal of KIAEBS, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Vol. 12, No. 6, pp.669-679, Dec. 2018.

[J1] ***Juwan Ha***, Kyungsoon Park, Hwanyong Kim, Younghak Song. "A Study of Zero Energy Building Verification with Measuring and Model-based Simulation in Exhibition Building", ARCHITECTURAL RESEARCH, Architectural Institute of Korea (AIK), Vol. 20, No. 3, pp.93-102, Sep. 2018.
Annual Best Paper

Conference Papers and Submissions

[C18] ***Juwan Ha***, Seungmin Lee, Yujin Kim, Jaeyoon Byun, Soolyeon Cho. "Beyond Static IT Loads: Service-Driven Dynamic Thermal Behavior of LLM Inference in AI Data Centers", 2027 ASHRAE Winter Conference, Chicago, IL, Jan. 23 – Jan. 27, 2027. (Under review)

[C17] Seungmin Lee, ***Juwan Ha***, Yujin Kim, Jaeyoon Byun, Soolyeon Cho. "Characterizing Operational Thermal Variability and Cooling Imbalance in GPU-Intensive AI Data Centers", 2027 ASHRAE Winter Conference, Chicago, IL, Jan. 23 – Jan. 27, 2027. (Under review)

[C16] Yujin Kim, ***Juwan Ha***, Seungmin Lee, Jaeyoon Byun, Soolyeon Cho. "Operational Data-Driven EnergyPlus Modeling of a Hybrid Cooling System for a High-Performance Computing Data Center", 2027 ASHRAE Winter Conference, Chicago, IL, Jan. 23 – Jan. 27, 2027. (Under review)

[C15] Jaeyoon Byun, ***Juwan Ha***, Seungmin Lee, Yujin Kim, Soolyeon Cho. "Thermal Load Interactions and Load Flexibility Potential in a Liquid-Cooled HPC Data Center – Summer Operational Analysis", 2027 ASHRAE Winter Conference, Chicago, IL, Jan. 23 – Jan. 27, 2027. (Under review)

[C14] ***Juwan Ha***, Ammar H.A. Dehwah, Karthikeya Devaprasad, Michael Poplawski. "A Whole-Building Demonstration of Differentiable Predictive Control with Stochastic Occupant Behavior", SimBuild 2026, Twelfth National Conference of IBPSA-USA, Minneapolis, MN, May 20-22, 2026.
Best Paper in Controls

[C13] Raja Bandari, ***Juwan Ha,*** Melisa Gold, Sanchana Bandari. "Integrating Urban Heat Island Effect and Solar Energy Potential at University Campus", 2026 ARCC-EAAE International Conference, April 8-11, 2026, Atlanta, Georgia, USA.

[C12] ***Juwan Ha***, Seungmin Lee, Yujin Kim, Soolyeon Cho. "Model Predictive Coordinated Control for Grid-Interactive Residential Building with Energy Recovery Ventilator, PV, and Battery Energy Storage System", 2026 ASHRAE Winter Conference, Las Vegas, NV, Jan. 31 – Feb. 04, 2026.

[C11] Seungmin Lee, ***Juwan Ha***, Yujin Kim, Soolyeon Cho, "Forecasting High-Impact Low-Probability Power Outage with Temporal Fusion Transformers: A Case Study Wake County", 2026 ASHRAE Winter Conference, Las Vegas, NV, Jan. 31 – Feb. 04, 2026

[C10] Yujin Kim, ***Juwan Ha***, Seungmin Lee, Soolyeon Cho, "Ontology-Based Semantic Modeling of HVAC Systems in the Office Building: A Foundation for Scalable Control and Digital twin Integration", 2026 ASHRAE Winter Conference, Las Vegas, NV, Jan. 31 – Feb. 04, 2026.

[C9] ***Juwan Ha***, Seungmin Lee, Yujin Kim, Soolyeon Cho, "Improving Energy and Data Efficiency in Factory Buildings: A Physics-Informed Neural Networks-Based Model Predictive Control (PINNs-Based MPC) Approach", 2025 ASHRAE Annual Conference, Phoenix, AZ, June 21 - 25, 2025.

[C8] Sangwoo Ham, ***Juwan Ha,*** Seungjae Lee**,** Donghun Kim, "Development of the AI-tech assistant to transfer research outcome knowledge for accelerating low-cost MPC technology deployment", 2025 ASHRAE Winter Conference, Orlando, FL, USA, February 10-12, 2025.

[C7] ***Juwan Ha***, Yujin Kim, Younghak Song. "A Study on the Verification Approach Temperature of Cooling Tower according to the Change L/G Ratio", Autumn Conference for Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Chungju, Republic of Korea, Nov. 2020.

[C6] Yujin Kim, ***Juwan Ha***, Younghak Song. "A Study on the Cooling Performance of Water-side Economizer System on The Chilled Water Return-side Applying Partial Operation", Autumn Conference for Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Chungju, Republic of Korea, Nov. 2020.
Best Paper Award of Fall Conference

[C5] Yujin Kim, ***Juwan Ha***, Younghak Song. "A Study on the Cooling Energy Consumption Cut-off in the Intermediate Season of the Data Center" Summer Conference for The Society of Air-conditioning and Refrigerating Engineers of Korea (SAREK), Pyeongchang, Republic of Korea, Jun. 2020.

[C4] Seongcheol Park, ***Juwan Ha***, Yujin Kim, Younghak Song. "A Study on the Thermal Storage Loss Rate for the Geothermal Storage System", Autumn Annual Conference of Architectural Institute of Korea (AIK), Daejeon, Republic of Korea, Oct. 2019.

[C3] ***Juwan Ha***, Younghak Song. "Comparison Analysis of Temperature and Humidity Index (THI) according to Ventilation Type of laying house", Autumn Conference for Busan-Ulsan-Gyeongnam Chapter of Architectural Institute of Korea (AIK), Jinju, Republic of Korea, Nov. 2018.

[C2] ***Juwan Ha***, Younghak Song. "THI Index within Laying house according to the Predicted Outside Temperature", Autumn Conference for Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Incheon, Republic of Korea, Nov. 2018.

[C1] ***Juwan Ha***, Younghak Song. "A Study on Economic Analysis for Adapting Ice Thermal Storage System to Retrofit of Heat Source System", Autumn Conference for Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS), Seoul, Republic of Korea, Nov. 2017.

Patents

[P3] "Operation Control System and Method for High Difference and Low Condensing Temperature Water on Cooling Towers", Patent application:10-2021-0048529. Patent registration: 10-2325379. Republic of Korea.

[P2] "The Notification System for Layer chicken", Patent application: 10-2019-0173894. Patent registration: 10-2363864. Republic of Korea

[P1] "Method for Predicting a Temperature-Humidity Index in a Windowless Laying Hen Houses Which Employing a Cross-Type or Tunnel Type Ventilation System", Patent application: 10-2019-0064580. Patent registration: 10-2308895. Republic of Korea.

Professional Services

Journal Reviewer

2026 – Present	Scientific Reports (x1)
2026 – Present	Energy and Buildings (x1)
2025 – Present	Energy Efficiency (x1)
2025 – Present	Solar Energy (x2)
2025 – Present	Journal of Building Engineering (x1)
2023 – Present	Energy and AI (x3)

ASHRAE Technical Paper Reviewer

2024 - Present	2027 ASHRAE Winter Conference (Chicago, IL) (x5)
	2026 ASHRAE Winter Conference (Las Vegas, NV) (x3)
	2025 ASHRAE Annual Conference (Phoenix, AZ) (x5)
	2025 ASHRAE Winter Conference (Orlando, FL) (x2)
	2024 ASHRAE Annual Conference (Indianapolis, IN) (x1)

REHVA HVAC World Congress Paper Reviewer

2025	CLIMA 2025: the 15th REHVA HVAC World Congress (Milan, Italy) (x1)

IBPSA Building Simulation (BS) Conference Reviewer

2026	2026 IBPSA-USA Simbuild Conference (Minneapolis, MN) (x5)
2025	2025 BS Conference (Brisbane, Australia) (x5)

Professional Affiliations

2026-Present	Full Member, Sigma Xi
2025-Present	Student Member, IBPSA-USA
2023-Present	Student Member, Association of Energy Engineers (AEE)
2023-Present	Student Member, International Institute of Building Enclosure Consultants (IIBEC)
2023-Present	Student Member, Association for Computing Machinery (ACM) SIGEnergy
2022-Present	Student Member, American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE)
2023-Present	Student Member, Korean American Scientists & Engineers Association (KSEA)
2018-Present	Member, International Building Performance Simulation Association (IBPSA)
2018-Present	Member, Architectural Institute of Korea (AIK)

2017-Present	Member, Korean Institute of Architectural Sustainable Environment and Building Systems (KIAEBS)

Other Activities

2024-2025	President, ASHRAE at NC STATE
2023-2024	Vice-President, ASHRAE at NC STATE
2023-2024	Secretary, PhD Student Association (PhDSA), Ph.D. in Design, NC STATE
2023-Present	ASHRAE Technical Committees (TC)
	TC 6.2: District Energy
	TC 7.5: Smart Building Systems
	TC 9.9: Mission Critical Facilities, Data Centers, Technology Spaces and Electronic Equipment

Research Proposal Contribution

2026	**DOE GENESIS: Phase 1 10-A, Securing U.S. Leadership in Data Centers – Data Center Load Flexibility (ITO)**
	AI-Orchestrated, Physics-Constrained Digital Twin for Modeling and Control of Multi-Timescale Data Center Load Flexibility
	Role: Graduate Student Contributor. Proposal submitted; not funded. PI: Dr. Ning Lu.
2024	**DOE Connected Communities 2.0**
	AIM-RESCUE: AI-Driven Management for Resilience and Energy in Sustainable Connected Communities Using Edge Technologies
	Role: Graduate Student Contributor. Proposal submitted; not funded. PI: Dr. Soolyeon Cho.
2024	**NSF AI-Ready Testbeds**
	SPARC: An AI-Ready Smart and Resilient Community Testbed for Power–Building–Transportation–Communication Nexus
	Role: Graduate Student Contributor. Proposal submitted; not funded. PI: Dr. Ning Lu.